BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: PR X (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Physics Paper–X (01) (21)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The Boolean equation for the NOR gate is.....

(A) X = A + B
(B) X = A+B overline
(C) X = AB
(D) X = AB overline

Q2. If magnetic field is applied parallel to the direction of electron beam, the electron will.....

(A) Deflect
(B) Not change its state
(C) Slow down
(D) Speed up

Q3. Telephone communication is the example of transmission of signals through.....

(A) Electric wires
(B) Optical fibers
(C) Electromagnetic waves
(D) All these

Q4. Which one or more of the three decay processes (α, β or γ) results in a new element?

(A) Only α
(B) Only β
(C) Only γ
(D) α and β

Q5. When combing our hair, we shift electrons from our hair onto the comb. The charge on our hair is.....

(A) Negative
(B) Zero
(C) Positive
(D) Infinite

Q6. The SI units for electrical power is:

(A) Ampere
(B) Coulomb
(C) Watt
(D) Volt

Q7. 1 kWh = ...

(A) 1000 J
(B) 0.36 hrs
(C) 3600 W
(D) 3.6×10^6 J

Q8. Which derived unit is equivalent to tesla ' T '?

(A) Nm/A
(B) NA/m
(C) N/Am
(D) Am/N

Q9. The waves produced by the oscillation of material particles are called:

(A) Mechanical waves
(B) Transverse waves
(C) Electromagnetic waves
(D) None of these

Q10. Various instruments sound different even when they play the same note due to difference in.....

(A) Loudness
(B) Pitch
(C) Intensity
(D) Quality

Q11. An external rearview car mirror is convex with radius of curvature 16.0 m. The focal length of the mirror is.....

(A) + 8.0 m
(B) − 8.0 m
(C) + 32.0 m
(D) − 32.0 m

Q12. Which of the given material will refract light more?

(A) Air
(B) Glass
(C) Diamond
(D) Water

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: For English Medium Students. Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. In which medium air or water, an echo is heard sooner? Why?

2. Suppose you stand on a swing instead of sitting on it. Will your frequency of oscillation increase or decrease?

3. An object is placed 30.0 cm in front of a converging lens of focal length 10.0 cm. Find the image distance and the magnification.

4. The sum of the charges on both the plates of a capacitor is zero. What does a capacitor store?

5. State Joul's Law and write the mathematical form.

6. A heating element on an electric range, operating on 240 V has resistance of 30.0 Ω. What current does it draw?

7. The primary coil of a transformer is connected to a DC battery. Is there an emf induced in the secondary coil? Why?

8. A portable X-ray unit has a step up transformer. The 220 V input of which is transformed to the 100 kV output. The primary coil has 50 loops, what is the number of loops in the secondary?

9. What are the Boolean equation to represent the output of NAND and NOR gates?

10. How information is different from data?

11. The atomic number of one particular isotope is equal to its mass number. Which isotope is it?

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: For English Medium Students. Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] What is the speed of sound?

(b) [3] Show that the mass spring system executes simple harmonic motion.

QIV.

(a) [4] What are spherical mirrors? How many types of spherical mirrors are there? Explain by giving an example.

(b) [3] A typical capacitor in an MP3 player has C = 0.10 µF. If a charge 5.0 µC is placed on the plates, what is the voltage across the capacitor?

QV.

(a) [4] What are ohmic and non ohmic devices? Sketch and interpret the VI characteristic graph to justify metallic conductor, filament lamp and thermistor as ohmic or non-ohmic materials.

(b) [3] A transformer connected to a 120 V AC line is to supply 9600 V for a neon sign. What is the ratio of secondary to primary turns?

QVI.

(a) [4] How is carbon-14 used to determine the ages of wood, bones and other artifacts?

(b) [3] Describe the operation of AND, OR logic gates by drawing their symbol and truth table.